Software Engineering Toolkit

Welcome!

Anvil

Why this course?

- Practical skills are important to your education
- In Summer, you'll build software for a customer as a team in **Software Project**
- From previous iterations, many students lacked hands-on experience
- This course gives you a **technical base** to build upon

What you'll learn

Skills we find relevant in our practice as engineers:

- Git & GitHub
- Linux
- Containers
- Backend APIs
- Web clients
- Mobile apps
- DBs
- Data pipelines
- ML basics

Course structure

10 labs taking you from understanding to building:

1. Products, git, and your skill
2. Fix and deploy a backend
3. RESTful backend, testing & DB
4. AI integration
5. Telegram bot
6. Data and web
7. ML recommender
8. Flutter app
9. Agentic AI
10. Hackathon

How to succeed

The course is designed to be **doable in the lab**.

- Attend the labs;
- Go through the tasks;
- Do extra if you have the skill;
- Ask questions from TAs, classmates, LLMs.

We want you to learn and apply on the go, this is the main skill.

Let's get started!